PARTE 1

$$\frac{dy}{dt} = y(1-y) \qquad y(0) = 0,1$$

$$\frac{dy}{y(1-y)} = dt$$

$$\int \frac{dy}{y(1-y)} = \int dt \rightarrow \frac{1}{y(1-y)} = \frac{A}{y} + \frac{B}{1-y} \rightarrow A - Ay + By = 1$$

$A = 1$

$B = 1$

$$\int \frac{1}{y} + \frac{1}{1-y} = t + c$$

$$\ln|y| - \ln|1-y| = t + c$$

$$\ln\left|\frac{y}{1-y}\right| = t + c$$

$$\frac{y}{1-y} = e^{t+c} \Rightarrow \frac{y}{1-y} = ce^{t} \rightarrow y = ce^{t} - ce^{t}y$$

$$y + ce^{t}y = ce^{t}$$

$$y(1 + ce^{t}) = ce^{t}$$

$$y = \frac{ce^{t}}{1+ce^{t}} \quad \div ce^{t} \rightarrow y = \frac{1}{1+Ke^{-t}}$$

$$\rightarrow y = \frac{1}{1+Ke^{-t}} \rightarrow y(0) = 0,1 \rightarrow 0,1 = \frac{1}{1+K.1} = \frac{1}{1+K} = 0,1$$

$$1 = 0,1 + 0,1K$$

$$0,9 = 0,1K$$

$$\boxed{K = 9} \longrightarrow \boxed{y = \frac{1}{1+9e^{-t}}}$$